

Lung deposition of fine and ultrafine particles outdoors and indoors during a cooking event and a no activity period.



Some indoor activities increase the number concentration of small particles and, hence, enhance the dose delivered to the lungs. The received particle dose indoors may exceed noticeably the dose from ambient air under routine in-house activities like cooking. In the present work, the internal dose by inhalation of ultrafine and fine particles is assessed, using an appropriate mechanistic model of lung deposition, accommodating aerosol, and inhalation dynamics. The analysis is based on size distribution measurements (10-350 nm) of indoor and outdoor aerosol number concentrations in a typical residence in Athens, Greece. Four different cases are examined, namely, a cooking event, a no activity period indoors and the equivalent time periods outdoors. When the cooking event (frying of bacon-eggs with a gas fire) occurred, the amount of deposited particles deep into the lung of an individual indoors exceeded by up to 10 times the amount received by an individual at the same time period outdoors. The fine particle deposition depends on the level of physical exertion and the hygroscopic properties of the inhaled aerosol. The dose is not found linearly dependant on the indoor/outdoor concentrations during the cooking event, whereas it is during the no activity period.

PRACTICAL IMPLICATIONS: The necessity for determining the dose in specific regions of the human lung, as well as the non-linear relationship between aerosol concentration and internal dose makes the application of dosimetry models important. Lung dose of fine and ultrafine particles, during a cooking event, is compared with the dose at no indoor activity and the dose received under outdoor exposure conditions. The dose is expressed in terms of number or surface of deposited particles. This permits to address the dosimetry of very small particles, which are released by many indoor sources but represent a slight fraction of the particulate matter mass. The enhancement of the internal dose resulting from fine and ultrafine particles generated during the cooking event vs. the dose when no indoor source is active is assessed. The results for those cases are also compared with the dose calculated for the measured aerosol outdoors.